Python Lists Notes

Based on Tea Varieties Practice Session

1. Creating a List

```
tea_varities = ['ginger', 'lemomn', 'masala', 'oolong']
```

- Lists use `square brackets []` — missing the opening `[` causes a SyntaxError
- Items are separated by commas and can be strings, numbers, or mixed types

💡 *Error encountered: SyntaxError: unmatched ']' — fixed by adding the opening [*

2. Accessing & Modifying Items

By Index

```
tea_varities[3] = "green"   # Replace item at index 3
# Result: ['ginger', 'lemomn', 'masala', 'green']
```

- Use `list[index]` to access or update a specific item
- Index starts at 0 — so index 3 is the 4th item

💡 *Error encountered: NameError: name 'green' is not defined — the value must be in quotes as a string*

By Slice

```
tea_varities[1:2] = ["green"]   # Replaces index 1 only
# Result: ['ginger', 'green', 'masala', 'green']

tea_varities[1:4] = ["blue"]    # Replaces index 1,2,3 with just one item
# Result: ['ginger', 'blue']
```

- Slice assignment `list[start:end] = [...]` replaces a range of items
- The replacement list can be shorter or longer than the slice — the list resizes automatically
- ⚠ `[1:4]` with one replacement item shrinks the list — be careful!

3. Looping Through a List

```
for tea in tea_varities:
    print(tea)
# Output: ginger / lemomn / masala / oolong (each on new line)
```

- The `for ... in ...` loop iterates over each item in the list
- The loop body MUST be indented (4 spaces or 1 tab)

💡 *Error encountered: IndentationError: expected an indented block — fixed by indenting the print() call inside the loop*

Custom Separator with end=

```
for tea in tea_varities:
    print(tea, end="-")
# Output: ginger-lemomn-masala-oolong-
```

- By default, `print()` adds a newline after each item
- Use `end="-"` to replace the newline with any character or string

4. List Methods

.append()

```
tea_varities.append("blue")
# Result: ['ginger', 'lemomn', 'masala', 'oolong', 'blue']
```

- `.append(value)` adds an item to the end of the list
- Pass the value inside the parentheses — `.append() = value` is wrong syntax

💡 *Error encountered: SyntaxError: cannot assign to function call — fixed by putting the value inside append("blue")*

.pop()

```
tea_varities.pop()
# Returns and removes the last item: 'blue'
```

- `.pop()` removes and returns the last item by default
- You can pass an index: `.pop(0)` removes the first item

5. range() and List Comprehension

range()

```
y = range(10)
# y = range(0, 10) — represents numbers 0 through 9
```

- `range(n)` generates numbers from 0 up to (but not including) n
- It doesn't create a list directly — it's a lazy sequence used in loops and comprehensions

List Comprehension

```
squared_num = [x**2 for x in range(10)]
# Output: [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

- Syntax: `[expression for item in iterable]`

- A compact way to build a list — equivalent to a for loop that appends each result
- `x**2` means x raised to the power of 2 (x squared)

6. Common Errors Encountered

`SyntaxError: unmatched ']'`
Opening [was missing. Always make sure both [and] are present when defining a list.

`NameError: name 'green' is not defined`
Value was written without quotes. Use "green" to pass a string, not green (which Python treats as a variable name).

`IndentationError: expected an indented block`
The body of a for loop must be indented. Press Tab or add 4 spaces before statements inside the loop.

`SyntaxError: cannot assign to function call`
append() = value is not valid syntax. The value goes inside the parentheses: append("blue").

7. Quick Reference

Operation	Example	Result
Create list	`['a', 'b', 'c']`	`New list`
Access item	`list[0]`	`First item`
Update item	`list[3] = "green"`	`Replaces index 3`
Slice replace	`list[1:3] = ["x"]`	`Replace range`
For loop	`for x in list:`	`Iterate items`
Custom end	`print(x, end="-")`	`No newline`
Append	`list.append("blue")`	`Add to end`
Pop	`list.pop()`	`Remove last`
Range	`range(10)`	`0 through 9`
Comprehension	`[x**2 for x in range(10)]`	`[0,1,4,9,...]`